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RESEARCH ARTICLE

Calculating and Visualizing Feature Importance Values for Counterfactual Explanations

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ABSTRACT Despite the success of machine learning algorithms, their opaque nature challenges responsible application. Counterfactual explanations have emerged as one potential solution to explain individual decision results, but their applicability is limited by 1) their isonomic view of feature changes, in which it is not possible to observe how much each modified feature influences the prediction, and 2) the lack of effective graphical resources to visualise the counterfactual explanation. We solve this by introducing Counterfactual Feature Importance (CFI) values. To calculate these values, we propose two methods: a greedy algorithm, and an adaptation of Shapley values to the feature changes of counterfactual explanations, which we term CounterShapley. We additionally introduce three chart types to visualise the counterfactual explanations: 1) the Greedy chart, which shows a greedy sequential path, 2) the CounterShapley chart, depicting its respective score in a simple one-dimensional chart, and 3) the Constellation chart, which shows all possible combinations of feature changes, and their impact on the model’s prediction score. For each of our proposed CFI methods and visualisation schemes, we demonstrate how they make counterfactual explanations more interpretable and actionable. To validate our approach, we surveyed 22 data scientists. 85.7% prefer at least one chart alongside counterfactual explanation. The majority of data scientists agree that the visualisations enhance the informativeness and usefulness of counterfactual explanations, and over 70% agree that visualisations increase their confidence in understanding counterfactual explanations. These findings suggest that counterfactual visualisations are an effective tool, highlighting their potential to support decision-making in data science contexts.

INDEX TERMS Counterfactual explanations, explainable AI, feature importance, visualization, XAI.

I. INTRODUCTION

AI models are susceptible to a trade-off between performance and complexity. If complex models cannot comprehensively justify their decisions, they are likely to pose significant ethical risks [1], [2]. This is already exemplified by contemporary models producing unfair or biased decisions based on erratic patterns in training data [3], [4], [5], or gender bias in Large Language Models [6]. Current legislation demands explainability from models that impact high-stake

decisions for people’s lives [7], and there have been multiple frameworks proposed to evaluate fairness in AI, such as AI Trust, Risk and Security Management (TRISM) [8]; the Fairness, Accountability, and Transparency (FAT) framework [4]; and multiple structured taxonomies [9], [10], [11]. Model explainability decisively impacts its applicability [5], both in high-stakes contexts where their decisions greatly affect human lives [7], [12], but also in common consumer-oriented applications, where it simply increases user satisfaction [13].

Within this context, the field of eXplainable Artificial Intelligence (XAI) dedicates great efforts to explain a Machine Learning (ML) model, prediction or decision.

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Many existing XAI methods include methods for calculating and visualising the feature importance (FI) values, or these FI values are the very explanation itself. Such visualisations greatly improve the end user's ability to understand and explain a piece of information [14], [15], [16]. However, visualising or quantifying the impact of individual features for the instance-based XAI method of CounterFactual Explanations (CFEs) is still lacking. This imposes a strong limitation on their interpretability and adoption. For these reasons, we introduce two methods to quantify Counterfactual Feature-change Importance (CFI) values, and three corresponding visualisations.

We summarize previous work in Section II and outline how this work compares, covering the motivation for CFI and visualizations in more detail. In Section III, we introduce a formal definition of counterfactual explanations. We use these definitions in Section IV to introduce CFIs using both a greedy algorithm, and Shapley values applied to counterfactual explanations, hereafter named "CounterShapley" to distinguish it from Shapley [17] or SHAP [18]. Section V visualises these CFIs using the Greedy chart, CounterShapley chart, and Constellation Chart. We highlight how these facilitate interpretability of CFEs in Section VI, including an example of how these can uncover hidden biases. Section VII covers a survey evaluating data scientists' opinions on these charts.

II. PREVIOUS WORK

We will briefly introduce existing methods of explaining a ML prediction or decision, and compare how they adhere to the following three qualitative metrics:

- 1) The **scope** can be local or global. Local explanation methods aim at uncovering why a model predicts some outcome for a single data instance, whereas global explanations seek to provide an overall explanation of how the model works on the entire dataset.
- 2) The **compatibility** of the explanation can be model-specific or model-agnostic. Model-specific methods often use particular mechanisms to generate explanations, inherent to the model inner workings. Model-agnostic explanation methods do not rely on this, and are applicable to any model.
- 3) The **output format** [5] of the explanations can be i.a. feature importance values, heatmaps for image explanations, or textual output for counterfactual explanations.

Popular global explanation approaches use intrinsic model characteristics, such as impurity decrease analysis for tree-based models [19]; rule extraction strategies, such as Trepan [20] and ALBA [21], where a less complex and more comprehensible surrogate model is built to mimic the closed box model; or depict the marginal effect of features on the prediction score, such as partial dependence plots (PDP) [22]. For local explanations, LIME [23] and SHAP [18] are popular feature importance (FI) methods, showing the most important features for a given instance's

prediction score. LIME considers datapoints close to the instance to be explained, and fits a locally linear relationship — a core idea that has been expanded to e.g. Graph Neural Networks [24], or to not assume independent features [25]. SHAP calculates approximate or exact Shapley values of individual features, compared to the average prediction and average feature value of the entire dataset. Counterfactual explanations (CFE) are instance-based explanations that aim to provide a set of feature value changes that would lead to a different model decision [26].

A. WHY COUNTERFACTUALS?

Ever since it has been introduced in 2011 [27], publications in the field of counterfactuals have been increasing [28]. As of today, over 300 counterfactual generation methods have been proposed [26], prioritising different properties of the CFE, including (but not limited to): stability [29], sparsity (defined below), feasibility and actionability [30], distance metrics between the factual and counterfactual instance, applicability for multiple instances at once [31], or to include uncertainty [32]. Sparsity is of particular importance in this work, as it has a significant impact on the algorithmic complexity of calculating Shapley values, which will be further discussed in Section IV-B.

The increased interest in counterfactual explanations can be understood by highlighting several desirable properties: they are simple, model-agnostic, sparse, and decision-oriented [26], [33], [34]. First, "simple" refers to the familiarity that humans have to use counterfactual arguments when providing explanations [35]; the output format is immediately interpretable. This process is well-established both in philosophy and psychology [35], [36], showing that such explanations come natural to end users. Second, counterfactual explanations do not rely on any specific model since they consider the prediction model merely in terms of input and output, and independent of its inner workings [28], making them model-agnostic and broadly applicable. Third, the explanations are typically sparse, meaning that only a small subset of all features occurs in the explanation. Not only does this make the explanation smaller and hence more comprehensible [4] (compared to an explanation using all features), but it also considerably reduces the complexity of the problem of assigning importance values to each feature change in the explanation, especially on the level of algorithmic complexity. Finally, the decision-oriented nature is a key point of CFE; they focus on modifications that lead to a relevant prediction change [34]. This characteristic is fundamental for highly complex machine learning methods, since their nonlinearity can lead to non-trivial feature-value relationships that can potentially bring misleading assumptions. We can verify this last point in literature works [34], [37] that compare counterfactual explanations with purely feature importance-based methods, and show that varying the "most important" features do not necessarily lead to a change in classification. This decision-oriented

nature has also been further explored in works such as XAI Stories [38], which proposed narrative-based approaches to improve the comprehensibility of AI decisions. Hence, CFE can be advantageous whenever one is more interested in explaining a decision, rather than a prediction score. This decision-oriented nature of counterfactuals aligns well with the transparency requirements in legislation, such as the European GDPR [7].

In addition to being used as stand-alone explanations, CFEs have also been used to derive FI values. For example, Mothilal et al. [39] relate the frequency with which a feature appears across multiple CFEs to its global FI, and Albini et al. [40] propose using a local sample of CFEs as a background dataset for SHAP to calculate instance-level FI values. We highlight these approaches to clarify the distinction between using CFEs to derive FI values and the present work, which focuses on calculating FI values for a single CFE itself.

B. WHY VISUALIZATION?

The importance of visualisation is not new in the field of counterfactual explanations [26]. They greatly improve the end user's ability to understand and explain a piece of information compared to text or tabular representations [14], [15], [16]. The commonly absent visualisations of CFEs is even a likely reason for consumers to favour feature importance methods as an explanation over textual counterfactual explanations [41].

However, visualisations do exist in the field of counterfactuals; it even traces back to its origins [27], where the greedy nature of the counterfactual generation algorithm (SEDC) allowed the plotting of sequential changes until the counterfactual class was achieved. GAMUT [42] implements a user interface that shows multiple statistics and methods to assess the relationship between the instance's features and the model's prediction. Google's What-If Tool [43] focuses on allowing the user to explore the model's behaviour by probing feature changes that are dynamically presented in tables and charts. DECE [44] also presents a user interface that includes various statistical analyses of features, and shows multiple counterfactual explanations according to sequential modification steps. VICE [45] represents counterfactual changes in charts together with the data distribution, and it allows customisation of which features to include in the explanation. Inspired by this, [46] proposed an interactive tool for tabular data, compatible with any counterfactual explanation generator. Reference [32] provides a qualitative and comprehensible visualisation of Counterfactual Explanations, in which they visualise probability estimates for the prediction score when singular features are changed [47].

While these visualisation methods exist, they are lacking, each for different reasons. First, not all CFE generators are greedy in nature like SEDC [27] is, and visualising CFEs can and should be improved upon to cover all possible types of CFE generators, not just greedy ones. Second, with the exception of [32] and SEDC [27], the additional

value of having a graphical interface or some form of visualisation is limited, and do not provide substantial insight beyond the bare CFE itself. These include visualisation methods representing counterfactual changes by simply showing which features were modified, or by showing spatial shifts in two-dimensional space, which lose a lot of information in the projection to 2D. Lastly, while [32] provides a clear and interpretable visualisation of CFEs, they mainly focused on providing uncertainty estimates from Calibrated Explanations. They only visualise CFEs created from those Calibrated Explanations, and only those CFEs where a single feature is changed. They can not generate or visualise CFEs where multiple features need to be changed at once, eliminating the possibility of investigating their interdependencies, and making it inapplicable to many other existing CFE generators.

The current and most widely adopted solution to represent a counterfactual explanation is often to provide a set of feature names and associated values to be changed in text format, as illustrated in Explanation 1.

Explanation 1: Rejection of a credit card application. If your age were 28 years old instead of 20, and your salary were \$4,000 instead of \$1,200 then you would be approved instead of rejected.

The aforementioned methods either do not provide substantial additional insight on the counterfactual explanation, the importance of the individual feature changes, or pose severe limitations in terms of applicability or user friendliness. For example, it is impossible to know how impactful certain feature changes are, or how their impact may depend on other feature changes. This emphasises the limitations CFEs have in their isonomic nature, the lack of methods to properly represent how each feature change impacts the model's prediction scoring, and lack of broadly applicable visualisation tools. For these reasons, we introduce Counterfactual Feature-change Importance (CFI) values that are applicable to all CFEs, can capture feature interdependencies, and lend themselves to easy-to-understand visualisations, without being dependent on any particular CFE generation algorithm.

To start, we will revisit the idea of a greedy visualisation of CFEs like SEDC [27] introduced, but we make it CFE generator agnostic in Sections IV-A and V-A. As these do not yet capture feature interdependencies, we will additionally introduce a way of assessing the impact that individual feature changes have on a ML decision in Section IV-B by adapting Shapley values to CFEs, hereafter referred to as CounterShapley values. After these We propose three visualisation methods in Section V: a greedy chart for the greedy CFIs, and CounterShapley and Constellation charts for the CounterShapley values.

III. COUNTERFACTUAL EXPLANATIONS

This section introduces a formal definition of a counterfactual explanation, which will return in Section IV to define CFI values. Counterfactual explanations assume three basic

components: a dataset ($\mathbf{D}$); an instance to explain ($\mathbf{x}$); and a prediction model ($\mathcal{M}$). For classification tasks, the model has an associated classifier ($\mathcal{C}$) that assigns instances to a class depending on the model's score and some threshold t . Given two different datapoints $\mathbf{a}$ and $\mathbf{b}$, we can define their difference as the set of indices where their feature values differ:

$$\delta_{a,b} = \{i \mid a_i \neq b_i\} \quad (1)$$

For notational ease, we define the scoring and classifying of the instance $\mathbf{a}$ with replaced feature values $\delta_{a,b}$ as:

$$\begin{aligned} \mathcal{M}(\delta_{a,b}) &= s \\ \mathcal{C}(\delta_{a,b}) &= c \end{aligned} \quad (2)$$

Note that the definition of δ is symmetric with respect to the two associated instances. By convention, this notation will denote the scoring of $\mathbf{a}$, adapted to have the values of $\mathbf{b}$ on the indices in δ , where $\mathbf{a}$ and $\mathbf{b}$ are the left and right instance of the $\neq$ sign in Equation 1. This becomes important as soon as we consider subsets of $\delta_{a,b}$. In the case of factual and counterfactual instances (see below), this notation will always denote the factual instance, adapted to have feature values of the counterfactual on some indices. Not the other way around.

Using this notation, we can formalise a counterfactual instance $\mathbf{c}$, derived from a factual instance $\mathbf{x} \in \mathbf{D}$, as an instance with the following properties:

- 1) The factual and counterfactual's predicted classes are not the same:

$$\mathcal{C}(\mathbf{x}) \neq \mathcal{C}(\mathbf{c}) \quad (3)$$

- 2) The difference between the factual and counterfactual can be defined by a set of indices where their features differ, i.e. the counterfactual explanation itself:

$$\delta_{x,c} = \{i \mid x_i \neq c_i\} \quad (4)$$

- 3) The counterfactual explanation is irreducible: there's no subset of valid changes that changes the original class.

$$\nexists \delta'_{x,c} \subset \delta_{x,c} : \mathcal{C}(\delta'_{x,c}) = \mathcal{C}(\delta_{x,c}) \quad (5)$$

As previously mentioned, there are a wide variety of counterfactual generation algorithms, prioritising different properties of the CFE. Sparsity $|\delta|$ is of particular importance in this work, as it has a significant impact on the algorithmic complexity of CounterShapley, which will be discussed in Section IV-B.

IV. FEATURE IMPORTANCE VALUES FOR COUNTERFACTUAL EXPLANATIONS

In this section, we introduce two methods of assigning feature importance values to counterfactual explanations: the greedy CFI method in Section IV-A, and CounterShapley values in Section IV-B. Both CFI methods only assume a factual instance, counterfactual instance, and a scoring function. They do not necessarily require a classifier, making them

applicable to regression tasks as well. They do not depend on any specific counterfactual explanation generator either, and are thus applicable to all of them.

A. GREEDY CFI METHOD

An obvious way of assigning feature importance values to feature changes is to change whatever feature change of the CFE has the biggest positive impact on the prediction score, and continue the same logic for the remaining feature changes i.e. a greedy search algorithm. This approach has several desirable properties:

- 1) The sum of the feature change scores is equal to the difference between the counterfactual and factual prediction ($\mathcal{M}(\mathbf{c}) - \mathcal{M}(\mathbf{x})$)
- 2) It has an intuitive interpretation that easily lends itself to visualisation (further discussed in Section V-A)
- 3) It is computationally efficient, with a complexity of $O(N \log N)$.
- 4) It is applicable when one wants to maximise the outcome for each feature change (e.g. when the classification threshold is unknown).

The downside of this approach is that this does not capture the impact a feature has on other features. Every single feature change in this greedy algorithm is made with a different baseline; each time, one more feature has been changed compared to the previous step. This makes it unclear what the individual impact of each feature is, and the interdependency of features and their impact on the model prediction is not at all captured with this method.

B. COUNTERSHAPLEY CFI METHOD

Given the general non-linear behaviour of complex models, different sequences of the same set of feature changes usually give different importance values. Therefore, we introduce the CounterShapley CFI method next, which presents a way to calculate importance values that consider all possible sequences, and are thus independent of a specific modification path. We will start by introducing Shapley values in general, follow up with a short discussion of the current use of Shapley in machine learning and why it needs adaptation in this specific case, formalise the application of Shapley to CFEs, briefly compare the core properties to Shapley and SHAP, and provide a general implementation overview.

1) SHAPLEY VALUES

Shapley values quantify the contribution of different players ($p \in Z$) to a final numeric result (φ). Each possible subset of players can form coalitions (V) that contribute to an outcome, defined by a gain function ($\mathcal{H}(V)$) [17]. Therefore, we can calculate the contribution (φ_i) of a certain player (p_i) in a coalition formed by a set of $|V|$ players by considering its marginal contribution in all possible coalitions:

$$\varphi_i(\mathcal{H}) = \sum_{V \subseteq Z \setminus \{i\}} \frac{|V|!(|Z| - |V| - 1)!}{|Z|!} (\mathcal{H}(V \cup \{i\}) - \mathcal{H}(V)) \quad (6)$$

These Shapley values have several desirable properties [48]. Most notable for the use case of CounterShapley values (see IV-B3) is efficiency: the sum of all players' contributions is equal to the total worth of the game or the grand coalition.

2) SHAPLEY VALUES IN MACHINE LEARNING

SHAP [49], [50] adapts Shapley values to a ML context by considering features to be players, the Shapley value as a feature importance value, and adapting the baseline prediction (an “empty” set of players) to be the average prediction score. It alleviates the issue of stability that LIME introduced [23], [51], where similar instances could yield vastly different feature importance values [51], [52]. SHAP provides model-agnostic approaches (e.g. KernelSHAP) as well as model-specific approaches (such as TreeSHAP [53]), and its scope spans from global explanations to local explanations, including local populations [54].

Shapley values and derived methods have not yet been applied to CFE. We therefore aim to define Counterfactual Feature change Importance (CFI) methods and corresponding visualisation resources that are tailored to the cause. Not to further explain some ML model, but to enhance the informative value of a pre-existing counterfactual explanation.

3) COUNTERSHAPLEY DEFINITION

Let us consider the counterfactual evidence set $\delta_{x,c}$ and associated feature values x_i and c_i , such that the factual feature values have to be changed from x_i to c_i in order to go from a factual $\mathbf{x}$ to a counterfactual $\mathbf{c}$. If we define the model $\mathcal{M}$ as the scoring function, $\delta_{x,c} = \delta$ and $|\delta| = K$ for notational ease, we can use Equation 6 to define a Shapley value φ_i for the counterfactual δ as:

$$\varphi_i(\mathcal{M}) = \sum_{V \subseteq \delta \setminus \{i\}} \frac{|V|! (K - |V| - 1)!}{K!} (\mathcal{M}(V \cup \{i\}) - \mathcal{M}(V)) \quad (7)$$

where K is the number of features, and the sum extends over all sets V that are a subset of $\delta \setminus \{i\}$. This weighted sum is simply the average marginal contribution of x_i to the class switch, as predicted by the model $\mathcal{M}$.

4) DIFFERENCE BETWEEN SHAPLEY, SHAP, AND COUNTERSHAPLEY

The definition of CounterShapley in Equation 7 may seem trivial compared to Shapley in Equation 6, but it has subtle and important differences to Shapley and SHAP, as summarised in Table 1. The most notable difference is the subscript of the summation sign, denoting which coalitions it is supposed to iterate. CounterShapley considers the feature value *changes* of $\delta_{x,c}$, rather than feature values, to be the players in Shapley. This is a core property of CounterShapley, and the necessary condition to make Shapley values applicable to CFE.

TABLE 1. Overview of the differences between Shapley values, SHAP, and CounterShapley.

	Shapley	KernelSHAP	CounterShapley
Baseline	$\mathcal{H}(\emptyset)$	$\sum_L \frac{\mathcal{M}(\mathbf{x}_i)}{L}$	$\mathcal{M}(\mathbf{x})$
Complexity	$\mathcal{O}(2^{ Z })$	$\mathcal{O}(L2^M)$	$\mathcal{O}(2^K)$ ($K \leq M$)
Efficiency	$\mathcal{H}(Z)$	$\mathcal{M}(\mathbf{x}_e) - \frac{\sum_L \mathcal{M}(\mathbf{x}_i)}{L}$	$\mathcal{M}(\mathbf{c}) - \mathcal{M}(\mathbf{x})$
Coalitions	Players Z	Sample $\mathbf{x}_i \in \mathbf{D}_{N,M}$	Counterfactual δ

5) IMPLEMENTATION OF COUNTERSHAPLEY

We finalise this section by describing how CounterShapley values can be calculated. In Section V, we also show how this implementation lends itself very nicely to visualisation purposes. Algorithm 1 presents a method of generating a map between any-sized coalitions of feature changes and their corresponding model prediction outcomes. This map can be used in conjunction with Algorithm 2 to calculate the Shapley value of the change in some feature i . Since Algorithm 1 has already concerned itself with mapping all possible coalitions to their corresponding model prediction outcome, Algorithm 2 does not need to perform any further calls to the model's prediction method and can simply fetch them from the result of Algorithm 1. This is an important aspect of not just this specific implementation, but any implementation, as this limits the number of calls to the model's prediction method. The extension to calculate all Shapley values for all features δ is now trivial, and can be done by using Algorithm 2 for every feature change $i \in \delta$. Calculating CounterShapley values is of complexity $\mathcal{O}(2^K)$, where K is not the total amount of features, but the amount of features changes present in the CFE i.e. $|\delta|$. As mentioned in Section III, CFEs can be optimized for sparsity, explicitly reducing K . This is a common objective in real-world applications, as high values for K directly impact the interpretability of CFEs. For reference, Explanation 1 has $K = 2$. In general, the runtime for calculating the Shapley values of a CFE is $\tau_{\mathcal{M}} 2^K$, where $\tau_{\mathcal{M}}$ is the duration of a single model inference. We have implemented multiprocessing for this algorithm, which divides this value by the amount of available CPUs (N_{CPU}). This algorithm is thus feasible as long as $\tau_{\mathcal{M}} 2^K / N_{CPU}$ is acceptable. A Python implementation of CounterShapley is available online at <https://github.com/ADMantwerp/CounterPlots>.

V. VISUALIZING COUNTERFACTUAL EXPLANATIONS

In this section, we present three different chart types that increase the informative value of counterfactual explanations. Two of them, Greedy and CounterShapley charts, are direct representations of the previously described CFI methods. The third chart type, the Constellation chart, is a different way of visualising CounterShapley values; rather

Algorithm 1 Generating Model Scores for Each Counterfactual Subcoalition

Input : factual ($\mathbf{x}$), counterfactual ($\mathbf{c}$), model ($\mathcal{M}$)
Output: $G = \{\text{coalition} : \text{score}\}$

```

1  $G \leftarrow \{\}$ 
2  $\delta \leftarrow \text{where}(\mathbf{x} \neq \mathbf{c})$  // indices of diff
3  $V \leftarrow \text{combinations}(\delta)$  // any-length
4 for  $V_i$  in  $V$  do
5    $\mathbf{c}^* \leftarrow \mathbf{x}$ 
6    $\mathbf{c}^*[V_i] \leftarrow \mathbf{c}[V_i]$ 
7    $G[V_i] \leftarrow \mathcal{M}(\mathbf{c}^*)$ 
8 end
9 return  $G$ 

```

Algorithm 2 Calculating the Shapley Value of Some Feature Change With Index i

Input: Feature index i , map G of all coalitions V and corresponding scores
Output: CounterShapley value φ_i of feature change i

```

1  $\varphi_i \leftarrow 0$ 
2 for  $V \subset \delta \setminus \{i\}$  do
3    $\text{score\_diff} \leftarrow G[V \cup \{i\}] - G[V]$ 
4    $\varphi_i \leftarrow \varphi_i + \text{score\_diff} \cdot L! \cdot (K - L - 1)! / K!$ 
5 end
6 return  $\varphi_i$ 

```

than just visualising the resulting CounterShapley values for each feature, it shows the model prediction score for all subcoalitions of a counterfactual explanation. It is precisely these subcoalitions that are iterated and weighted when calculating Shapley values. To exemplify their use, we will consider the Counterfactual Explanation 2, provide all three chart types for this counterfactual explanation, briefly describe how to interpret them, and highlight exactly which additional insights are given by each chart type. All visualisation tools mentioned below are available online at <https://github.com/ADMAntwerp/CounterPlots>.

Explanation 2: A loan application decision. If your age were 30 instead of 20, salary were \$2,200 instead of \$1,500, sex were F instead of M, and state were CA instead of NY, the loan application would be classified as approved instead of rejected.

A. GREEDY CHART

The greedy chart aims to illustrate a sequential, easily interpretable, and low-complexity visualisation of the greedy CFI.

1) INTERPRETATION

Figure 1 depicts how Explanation 2 would be displayed using the Greedy chart. The first point at the bottom is the baseline, and represents the prediction score of the factual point, which has the features *Sex*, *City*, *Salary*, and *Age* equal

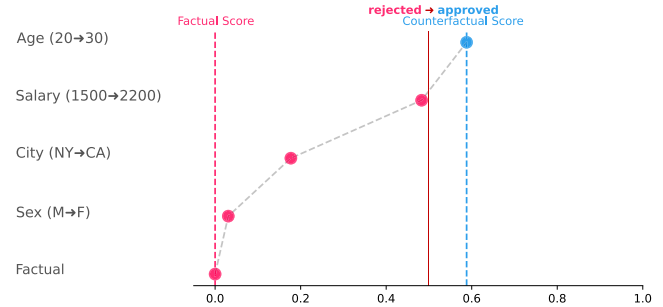


FIGURE 1. CFE visualisation for Explanation 2 using the Greedy chart.

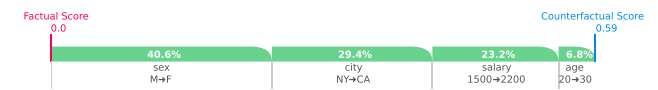


FIGURE 2. CFE visualisation for Explanation 2 using CounterShapley chart.

to “M”, “NY”, 1, 500, and 20 respectively. The next point shows the change in prediction score when the feature *Sex* is changed from “M” to “F”; this modification is the one that leads the highest increase towards the counterfactual class compared to the previous point (the baseline, in this case). Each consecutive point now reflects the feature change (note the y-labels) with the largest effect, compared to the relative baseline of the previous point.

2) INSIGHTS

Changing the “City” from “NY” to “CA” has a larger impact on the model score than changing “Sex” from “M” to “F”, but is not the first change in line. This is because the former change would not be as influential if the “Sex” wasn’t changed first. This is exactly the nonlinear nature of the model. Therefore, practitioners must be aware of whether this chart meets their expectations in explaining a decision, and possibly avoid it if they want to provide a precise estimation of feature change impact over the prediction score of complex models. This chart is appropriate when one wants to show the most influential feature to change “right now” This can be useful when the decision threshold is unknown, or the amount of actions one can undertake is limited.

B. COUNTERSHAPLEY CHART

CounterShapley values allow us to create a new visualisation, here referred to simply as the CounterShapley chart. Our main objective with this chart is to represent CFE in a comprehensive way, including the nonlinear nature of feature changes. This is a direct visualisation of the output of Algorithm 2.

1) INTERPRETATION

Figure 2 shows the CounterShapley chart for Explanation 2, summarising diverse information in a single image. The chart shows the original instance’s score in red on the left, and the counterfactual score in blue on the right. The green bars

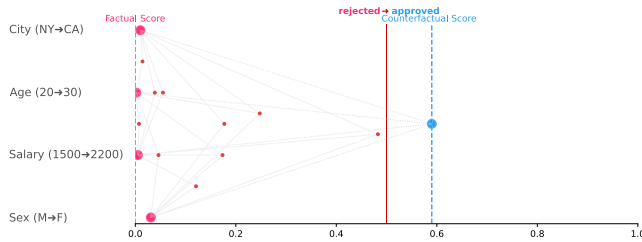


FIGURE 3. CFE visualisation for Explanation 2 using the constellation chart.

then represent the CounterShapley values for each feature change, sorted from high to low, visualising the proportional contribution each feature has to this score difference.

2) INSIGHTS

It's remarkable to note how the Greedy chart tells us that changing "Sex" does not increase the score by a lot initially, while the CounterShapley chart tells us that "Sex" has the biggest effect on the prediction score when considering all possible combinations of feature changes. "Salary" is almost the least influential feature change, in stark contrast with Figure 1. The large impact of "Salary" in the Greedy chart is thus *only* because the previous feature changes allowed it to make as big of an impact. Even though "Sex" does not appear to have a large impact initially (see Figure 1), its large CounterShapley value here should raise an eyebrow. This will be further discussed in the context of catalyst features in Section V-C. This chart is applicable when one wants to show the impact of each feature change, in those cases where it is certain that all features will indeed be changed. This is e.g. applicable when comparing two non-synthetic data instances.

C. CONSTELLATION CHART

Counterfactual explanations tend to have sparse feature changes. As discussed in Section IV-B2, this yields unique visualisation opportunities. The Constellation chart makes good use of this characteristic by displaying how every combination of feature changes influences the counterfactual prediction score. Rather than visualising the average marginal contribution (the output of Algorithm 2), it untangles all possible subcoalitions of feature changes and how they impact the model prediction score, which is exactly the output of Algorithm 1. For larger sets of feature changes, this chart does become unwieldy. For this reason, we also provide an interactive constellation chart, where hovering subcoalitions shows the adapted features and their prediction score; and clicking subcoalitions highlights only the selected subcoalition and its connecting lines with the relevant feature changes. In addition, we allow the user to pass a feature change index as keyword argument for the static constellation plots, so that only those coalitions containing this feature change are highlighted. These can be explored in our tutorials provided at <https://github.com/ADMAntwerp/CounterPlots>.

1) INTERPRETATION

The Constellation chart in Figure 3 shows how each single feature change ("Age", "Salary", "City", and "Sex") and how all their possible combinations affect the prediction score. The first pink dashed line on the left shows the factual score, as measured on the x-axis. The four large pink dots represent a single feature change. For example, the bottom large pink dot indicates the prediction score when the feature "Sex" is changed from "M" to "F". The smaller pink dots represent combinations of different combinations of feature changes, also visible from the grey lines connecting to the singular feature changes. The rightmost point represents the prediction score when all feature changes are applied: the counterfactual score.

2) INSIGHTS

Although this view looks more complex than the other previous methods, it also allows an extensive analysis of the influence of all feature changes. For example, this chart already shows how the exclusion of "Age" from the counterfactual still yields an instance that's incredibly close to the decision threshold; an insight that was absent from the previous two charts.

"Sex" is here fully revealed to be a catalyst feature; it does not directly impact the prediction score a lot on its own, but it does allow for other features to make a bigger impact: all subcoalitions (small pink dots) that include sex (connected with grey line) have a systematic substantially higher impact on the model prediction compared to those coalitions that do not include "Sex". This also explains its high CounterShapley value in Figure 2. Because of the low direct impact on the outcome of the model, catalyst features can be extremely difficult to spot. However, they are completely exposed here. This chart can therefore be an applicable tool for uncovering implicit unfairness in ML models: it provides a more exhaustive assessment of the impact of socially charged catalyst features.

This chart is applicable when one wants to show the impact of each feature change, across all possible combinations of changes. This can be useful for investigating the inner workings and biases of the prediction model. The Constellation Chart can also serve as a debugging tool for counterfactual explanations, since it is visually obvious if any subset of changes also crosses the decision threshold (see property 3 of the definition of counterfactuals in Section III), which will be explicitly shown in Section VI-C.

VI. RESULTS

In addition to improving upon the shortcomings of counterfactual explanations, the methods introduced here also unlock new ways of probing both the CFE generator, as well as the ML model. In this section, we will show how CFIs and the proposed visualisations can improve our understanding of some non-intuitive characteristics of CFE, probe ML model performance, and visualise false counterfactual

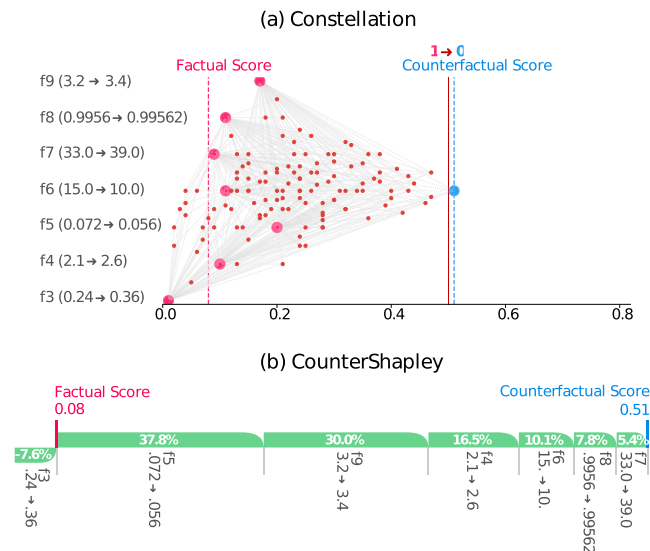


FIGURE 4. Visualisation of a counterfactual containing a feature with a negative CounterShapley value using (a) a Constellation chart and (b) a CounterShapley chart. Using the Wine dataset [58] and NICE [55] CFE generator.

explanations. We will generate counterfactual explanations using NICE [55] and DiCE [56], but the results work equally well for other counterfactual methods, such as ALIBIC [57] or SEDC [34]. Datasets for Section VI-A, Section VI-C and Section VI-B are UCI ML's wine dataset [58], a Bankruptcy Prediction dataset [59], and UCI ML's Adult Census Income dataset [60] respectively. Models are created with `sklearn` using default parameters. Explicit details on the datasets, ML model creation, and reproductions of the same graphs are available at <https://github.com/ADMAntwerp/CounterPlots>. The variety of both CFE generator algorithms and datasets is intended to highlight the applicability of our methods.

A. NEGATIVE COUNTERSHAPLEY VALUES

The impact of features in non-linear ML models usually depends on other features. As shown in Figure 4, our analysis can detect features that have an average negative effect on prediction scoring in certain feature coalitions, but are simultaneously necessary conditions for a class flip in other coalitions. In Figure 4, we show one example instance that has a feature (f_3) with a negative impact on the model score, and even a negative CounterShapley value, as seen in its respective CounterShapley chart. This means that this feature has a negative effect, not only when it is the only one changed, but also when averaged over all subcoalitions. The Constellation chart shows that f_3 decreases the class probability if changed alone (bottom clear blue dot), and even in association with many other features. However, as can also be seen in the Greedy and Constellation chart, this feature is fundamental for the class change. Within the context of other feature changes, it remarkably acquires a positive effect.

These charts highlight how a CFE should not necessarily be mistrusted even if it includes feature changes that have,

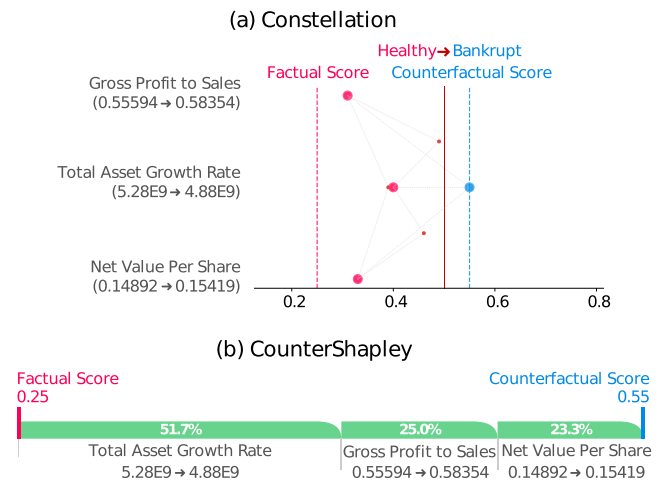


FIGURE 5. Visualisation of a counterfactual whose factual instance was incorrectly classified by the model as a healthy company (but it got bankrupt instead) using (a) a Constellation chart and (b) a CounterShapley chart. Using the Bankruptcy prediction dataset [59] and NICE [55] CFE generator.

on average, a negative impact on the prediction score. It is perfectly possible for CFEs to accurately propose feature changes that usually have a negative effect on the prediction score.

B. MISCLASSIFICATION

One of the most valuable uses of explanation methods, especially those that focus on instance-level explanations, is the investigation of why a model assigned a wrong classification to a given instance. If one is to trust the prediction of a ML model, surely the prediction should be accurate. CFE, enhanced with the aforementioned CFI methods, can reveal useful patterns and features to be further investigated if one wants to probe why a model is making wrong predictions.

Figure 5 shows the CFI values for an instance that was classified as a healthy company, but in reality got bankrupt. It is now of interest to ask the counterfactual question “What is the most similar situation that would have been classified as bankrupt?” This could provide us with information on which features the model is inaccurately capturing, or how close the model was to a correct prediction for this instance, effectively aiding us in improving the model accuracy.

The Constellation Chart shows how, if the total asset growth rate was 7.5% lower, gross profit sales was 5.0% higher, and net value per share 3.5% higher, the model would have made a correct classification. The first feature change is relatively sensible, since a slower growth is generally seen as a pessimistic indicator of a company's prospects, nudging the classification closer to an accurate prediction of bankrupt. The latter two changes look unexpected however, since both increases are associated with positive outcomes. For this reason, it's important to notice that these two changes alone are *not* sufficient to flip the classification and *must* be in

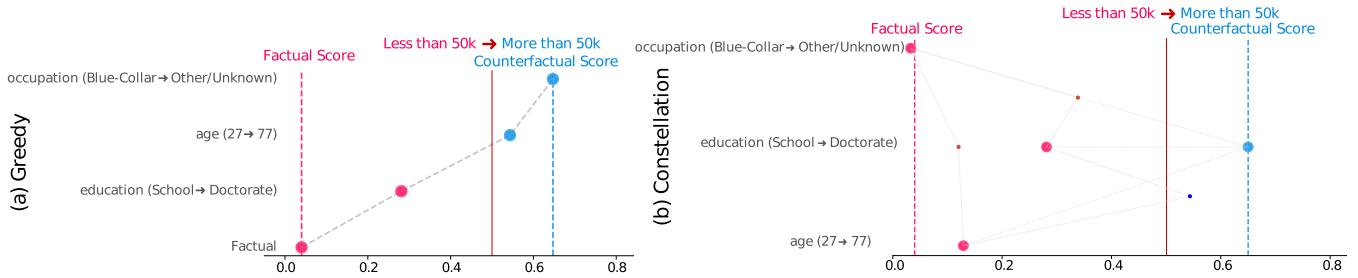


FIGURE 6. Visualisation of a counterfactual with a subset that also flips the prediction class using (a) a Greedy chart and (b) a Constellation chart. Using the Adult dataset [60] and DiCE [56] CFE generator.

conjunction with the first, which is also the most influential feature change.

Why would these changes lead to a correct prediction by the model, but the original instance did not? This requires further investigation, but thanks to this analysis, it can be much more targeted. Maybe the combination of higher shares and profit, together with a lower asset growth is not sustainable in the long run? One can explore if there are any similar instances in the dataset, or if these features correlate similarly across other instances too. Maybe the model is simply inaccurate, and this instance would have been classified correctly if the model was trained better? Or perhaps the real reason for bankruptcy can not be captured with the available data, in which case no XAI method would help.

The counterfactual explanations associated with the CFI methods and visualisations give us various targeted directions of inquiry. Using CFI methods as a CFE debugging tool, one can probe ML models in novel ways, and improve upon both the CFE generator algorithm, as well as the ML model itself.

C. COUNTERFACTUAL SUBSET

The definition of a counterfactual explanation includes the requirement of having an irreducible set of valid modifications (as described in Equation 5). Although the interpretation of valid modifications is debatable [33], identifying subsets of a CFE that also form a CFE may improve our understanding of both the prediction model, as well as the counterfactual explanation itself.

Figure 6 shows how visually obvious these subsets are in a Constellation chart. In addition to showing false CFEs, it also shows if certain subcoalitions of CFEs are close to the decision boundary (see Figure 3). One could make minor variations in the feature changes that lead to an observed near-counterfactual and check if any of them lead to a prediction score beyond the decision threshold. The Constellation chart visualises which feature changes are worth tinkering around with. This kind of visualisations provides insight in the quality of the CFE and its associated CFE generator, but also enables human analysts to identify and explore alternative CFEs that the CFE generator may have missed. In contrast to Constellation charts, Greedy

charts have no guarantee of showing a false CFE; the particular one in Figure 6 just happens to be visible in the Greedy Chart as well. The CounterShapley chart has been omitted, as subsets are not obvious there.

VII. SURVEY

A. SURVEY DESIGN

We conduct a survey that evaluates data scientists' opinions regarding three charts. The survey targets an audience of data science practitioners, as they already have an understanding of model prediction scoring, visualisation techniques, and the subtleties of machine learning non-linearities. We considered such a target audience to be ideal to evaluate our proposed methods.

Participants are informed of the survey outline beforehand and may consent to participate or not. Participants are presented with three different charts for one CFE, each accompanied by a detailed explanation of its interpretation, following with a question to check their understanding of the chart. The time taken to answer each question has been recorded to exclude possible inattentive responses. After all three questions are answered, participants are again shown the CFE along with its three charts. They are then asked whether they would use at least one chart, as well as to indicate their preferences among the three charts. Finally, participants rate their agreement with five dimension statements on a 5-point Likert scale (similar to [61]), ranging from 1 (*strongly disagree*) to 5 (*strongly agree*):

- 1) **Useful addition:** I think the visualisations are a useful addition to the counterfactual explanations for explaining the model's prediction.
- 2) **Ease:** I feel the visualisations are easy to interpret.
- 3) **Confidence:** I feel more confident in my understanding of the prediction with the visualisation than without the visualisation.
- 4) **Speed:** I feel I am able to arrive more rapidly at an understanding of the prediction with the visualisation than without the visualisation.
- 5) **Informative value:** I believe visualisations enhance the informative value of counterfactual explanations.

To examine whether the number of counterfactual features influences participants' opinions, we create two versions of the survey: version one includes charts with three

TABLE 2. Comparison of visualisation preferences. Group 1 views charts with three counterfactual feature changes, whereas Group 2 are shown charts with four counterfactual feature changes. We identify a significant difference in the distribution of preferences between the two groups: $\chi^2(2, 22) = 7.8, p = .048$.

Chart	Overall	Group 1 $ \delta = 3$	Group 2 $ \delta = 4$
Greedy	14.3%	11.1%	16.7%
CounterShapley	28.6%	0.0%	50.0%
Constellation	52.4%	77.8%	33.3%
No preference	4.8%	11.1%	0.0%
I don't like any	0.0%	0.0%	0.0%

counterfactual features, and version two includes charts with four. Participants are randomly assigned to one of the two versions, with those in version one categorized as Group 1 and the other Group 2. Each version includes two sub-versions, each featuring a distinct counterfactual explanation (CFE). All four CFEs and charts are generated using the Adult Census Income dataset [60]. Responses are collected from authors' research community, excluding members of the research group.

B. PARTICIPANTS' PREFERENCE RESULTS

When participants are asked: "As a data scientist, do you prefer using only counterfactual explanations, or including at least one chart along with the counterfactuals?" A strong acceptance of CFE visualisations among participants is received, with 85.7% choosing to include at least one chart. Group 1 shows a slightly higher preference (88.9%) than Group 2 (83.3%); however, the differences are not statistically significant, $\chi^2(2, 22) = 2.85, p = .240$.

Table 2 illustrates participants' preferences for a specific chart, with darker green shades representing higher percentages of selections. The Constellation chart emerges as the most preferred (52.4%) overall, with the CounterShapley chart 28.6%. Distinct patterns are revealed between the two groups: the majority (77.8%) of Group one favours the Constellation chart, whereas Group two exhibits a more varied distribution of preferences, with 50.0% selecting the CounterShapley chart and only 33.3% for the Constellation chart, $\chi^2(2, 22) = 7.88, p = .048$. Specifically, participants who view charts with three counterfactual features show a preference for the Constellation chart, whereas those who view charts with four counterfactual features prefer the CounterShapley chart. This result suggests that preferences for the Constellation chart decrease as the number of counterfactual features increases, likely due to the drastically increasing complexity of the Constellation chart.

C. FIVE DIMENSION RESULTS

Table 3 summarises the evaluation of the five user experience dimensions, showing the percentage of participants who select "Agree" or "Strongly Agree." A one-sided t-test is conducted to assess whether each percentage exceeds the 50% baseline, with statistically significant results marked

TABLE 3. Comparison of agreement on five dimension statements. Group 1 viewed charts with three counterfactual feature changes, whereas Group 2 were shown charts with four counterfactual feature changes. Each value in the table represents the proportion of participants who selected either "Agree" or "Strongly Agree" for each statement. A one-sided t-test reveals if each value is statistically significantly higher than 50%: *** $p < .001$, ** $p < .01$, * $p < .05$.

Dimension	Overall	Group 1 $ \delta = 3$	Group 2 $ \delta = 4$
Useful addition	90.5%***	100.0%***	83.3%**
Ease	52.4%	44.0%	58.3%
Confidence	71.4%*	66.7%	75.0%*
Speed	57.1%	44.4%	66.7%
Informative value	90.5%***	88.9%**	91.7%***

by asterisks. The results demonstrate consistency between the overall and group-specific analyses, revealing higher percentages of agreement for the dimensions of "useful addition," "informative value," and "confidence," whereas lower percentages are observed for the improvements in ease and speed of comprehension. Overall, the proportion of participants who agree that CFE visualisations are a useful addition and enhance the informative value of counterfactual explanations is significantly more than chance (90.5%, $p < .001$). The improvement in confidence is also significant, even with a slightly lower percentage (71.4%, $p = .023$). The results show a non-significant improvement in speed (57.1%, $p = .263$) and ease (52.4%, $p = .417$). Trends in both Group one and Group two closely mirror those observed in the overall, with the only notable difference being that the improvements in confidence are not statistically significant in Group one (66.7%, $p = .173$).

Additionally, a difference-in-proportion test is conducted for each dimension to examine whether the differences across the two groups are statistically significant. The results reveal no statistically significant differences, suggesting that the number of counterfactual features does not significantly influence participants' opinions across any of the five dimensions.

While several results reach statistical significance, the relatively small sample size and the specific participant pool limit the generalizability of the findings. Therefore, the results should be interpreted as indicative rather than broadly applicable.

VIII. CONCLUSION

Machine learning predictions can be effectively explained with counterfactual explanations. Its simple and decision-driven nature makes it easier for even laypersons to understand why certain automated decisions are made. Until now, the common representation of counterfactual explanations was purely textual, which lacks the interpretability that graphical elements provide. Moreover, the isonomic view of the counterfactual modified features does not give a comprehensive picture of why a model makes certain decisions, as it does not show the effect of each feature change on the model score.

This paper proposes two methods of assessing counterfactual feature importance (CFI) values as a solution: a CFE-specific greedy algorithm, and an interpretation of Shapley values for CFE, coined CounterShapley. These CFI methods now allow for new visualisation possibilities, which vastly increase the interpretability and applicability of counterfactual explanations. CounterShapley charts directly visualise the CFI values to create simple, yet descriptive graphs; the Greedy chart provides a visual representation of a greedy search through counterfactual feature changes; and the Constellation chart takes advantage of the sparse nature of counterfactual explanations to show all possible subcoalitions of counterfactual feature changes, and their impact on the prediction score.

The survey results reveal a statistically significant positive attitude towards our visualisations, within our sample of data science practitioners. Most participants prefer incorporating at least one chart alongside textual counterfactuals, reflecting their widespread acceptance of visualisations. The findings highlight the contribution of the three charts in enhancing the usefulness and informativeness, as well as boosting participants' confidence in understanding CFEs. However, they do not statistically improve the speed or ease of understanding. This outcome is expected, as the charts contain elements that demand a certain level of cognitive effort to interpret. Moreover, the Constellation chart emerges as the most preferred, particularly among participants presented with fewer counterfactual features, whereas preferences shift toward the CounterShapley chart as the number of features increases.

This work is presented as a proof-of-concept, which implies some limitations. First, we have not tested the applicability of these methods in real-world scenarios or beyond academic datasets. This is a common issue in data science, and we assume the proposed methods to be applicable in real-world scenarios as well, including large datasets, various CFE generators, and complex ML models. They scale well with large datasets, because counterfactual explanations do not depend on each other, and can easily be generated in parallel; they are sparse and generally iterate small subsets of feature value changes, in stark contrast to e.g. SHAP, which does iterate all features; and the CFE generator can always be chosen based on an efficient stopping criterion, in case CFE generation time provides a bottleneck. Furthermore, the proposed methods only assume that a factual, counterfactual, and scoring function are given, which makes them compatible with any CFE generator or ML model, independent of their complexity.

A second limitation to highlight is the exponential runtime of Shapley value generation. While we have made efforts to increase performance as much as possible, and CFE generators often prioritize low-dimensional CFEs, there are contexts in which it is possible for CFEs to become large-dimensional. If the model inference duration is computationally expensive as well, these contexts become less suitable for

the calculation of exact CounterShapley values. Nonetheless, for common ML models and even for unreasonably sized CFEs up to 10 dimensions, we have found the runtime to be near-instantaneous, ranging from a handful of milliseconds for regular-sized CFEs to sub-second for 10-dimensional CFEs. Performance should not be an issue for common use cases.

The third limitation of this study arises from the survey. The charts are generated from a single database containing only four CFEs, divided into two groups for comparison, and the analysis is based on just 22 valid responses. In addition, participants are asked to evaluate five dimensions based on their impression of three charts collectively, which may have obscured potential variations in the performance of an individual chart. Addressing these limitations in future studies could lead to more robust and comprehensive survey findings.

We envision that the application of counterfactual-based feature importance scores and their associated visual representations will greatly promote the use of CFE. In conjunction with their intuitive interpretation, these enhancements may even make CFE the preferred way of explaining complex AI model decisions to end users.

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